



EXAM PAPERS PRACTICE

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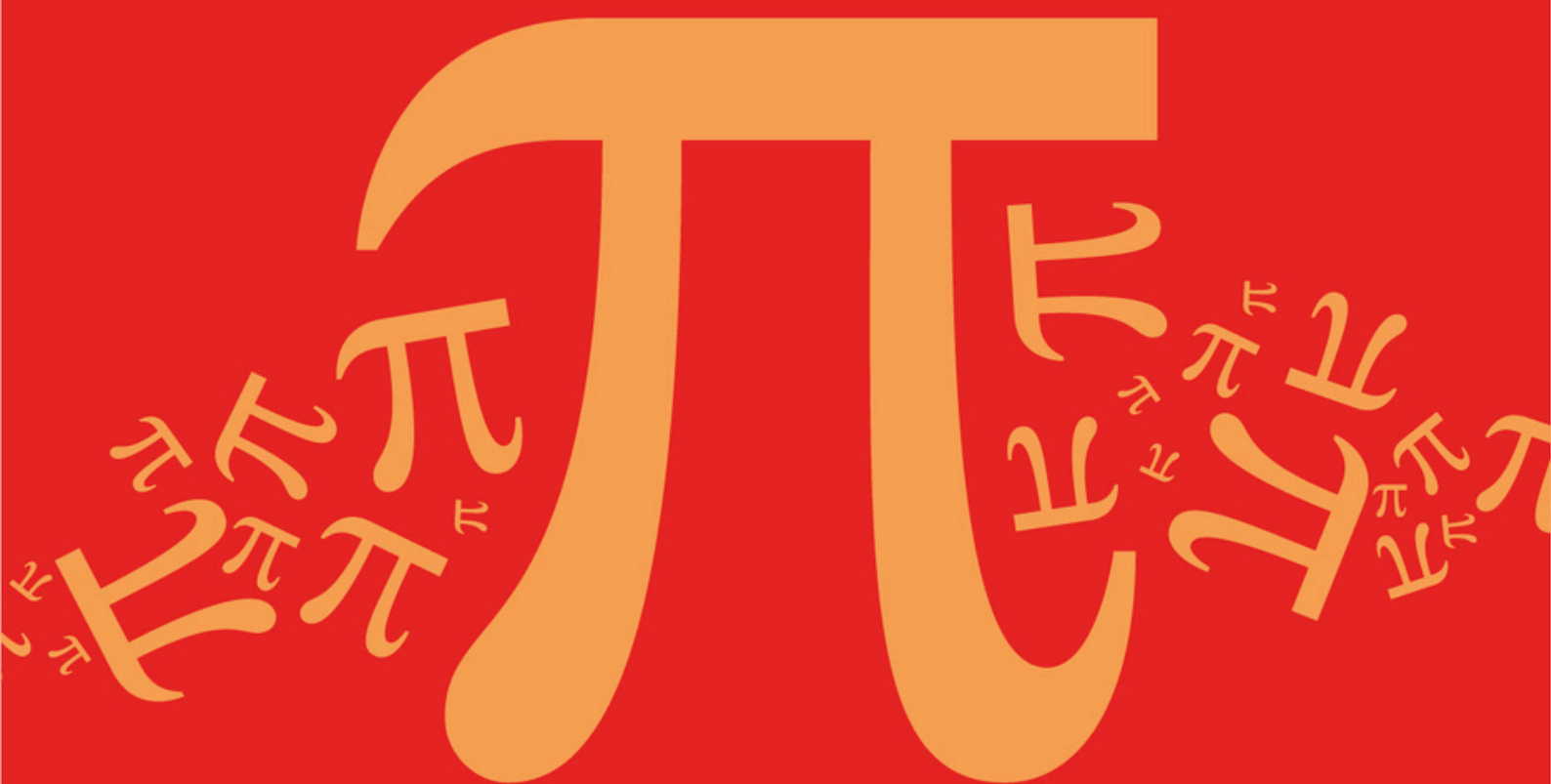
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AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B8: De Moivre's Theorem

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

B: Complex Numbers

Sub topic

B8: De Moivre's Theorem

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

Find the smallest value of θ for which

$$(\cos \theta + i \sin \theta)^5 = \frac{1}{\sqrt{2}}(1 - i) \quad \{\theta \in \mathbb{R} : \theta > 0\}$$

(Total 4 marks)

Q2.

- (a) (i) Use de Moivre's theorem to show that

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

and find a similar expression for $\sin 4\theta$.

(5)

- (ii) Deduce that

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

(3)

- (b) Explain why $t = \tan \frac{\pi}{16}$ is a root of the equation

$$t^4 + 4t^3 - 6t^2 - 4t + 1 = 0$$

and write down the three other roots in trigonometric form.

(4)

- (c) Hence show that

$$\tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{7\pi}{16} = 28$$

(5)

(Total 17 marks)

Q3.

Find the smallest positive integer values of p and q for which

$$\frac{\left(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8}\right)^p}{\left(\cos \frac{\pi}{12} - i \sin \frac{\pi}{12}\right)^q} = i$$

(Total 7 marks)

Q4.

- (a) Write down the five roots of the equation $z^5 = 1$, giving your answers in the form $e^{i\theta}$, where $-\pi < \theta \leq \pi$.

(1)

- (b) Hence find the four linear factors of

$$z^4 + z^3 + z^2 + z + 1$$

(3)

- (c) Deduce that

$$z^2 + z + 1 + z^{-1} + z^{-2} = \left(z - 2 \cos \frac{2\pi}{5} + z^{-1} \right) \left(z - 2 \cos \frac{4\pi}{5} + z^{-1} \right)$$

(4)

- (d) Use the substitution $z + z^{-1} = w$ to show that $\cos \frac{2\pi}{5} = \frac{\sqrt{5} - 1}{4}$.

(6)

(Total 14 marks)

Q5.

- (a) Use De Moivre's Theorem to show that, if $z = \cos \theta + i \sin \theta$, then

$$z^n + \frac{1}{z^n} = 2 \cos n\theta$$

(3)

- (b) (i) Expand $\left(z^2 + \frac{1}{z^2} \right)^4$.

(1)

- (ii) Show that

$$\cos^4 2\theta = A \cos 8\theta + B \cos 4\theta + C$$

where A , B and C are rational numbers.

(4)

- (c) Hence solve the equation

$$8 \cos^4 2\theta = \cos 8\theta + 5$$

for $0 \leq \theta \leq \pi$, giving each solution in the form $k\pi$.

(3)

- (d) Show that

$$\int_0^{\frac{\pi}{2}} \cos^4 2\theta \, d\theta = \frac{3\pi}{16}$$

(3)

(Total 14 marks)

Q6.

- (a) (i) Use de Moivre's Theorem to show that

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

and find a similar expression for $\sin 5\theta$

(5)

- (ii) Deduce that

$$\tan 5\theta = \frac{\tan \theta (5 - 10 \tan^2 \theta + \tan^4 \theta)}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

(3)

- (b) Explain why $t = \tan \frac{\pi}{5}$ is a root of the equation

$$t^4 - 10t^2 + 5 = 0$$

and write down the three other roots of this equation in trigonometrical form.

(3)

- (c) Deduce that

$$\tan \frac{\pi}{5} \tan \frac{2\pi}{5} = \sqrt{5}$$

(5)

(Total 16 marks)

Q7.

The cubic equation

$$2z^3 + pz^2 + qz + 16 = 0$$

where p and q are real, has roots α , β and γ .

It is given that $\alpha = 2 + 2\sqrt{3}i$.

- (a) (i) Write down another root, β , of the equation.

(1)

- (ii) Find the third root, γ .

(3)

- (iii) Find the values of p and q .

(3)

- (b) (i) Express α in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(2)

- (ii) Show that

$$(2 + 2\sqrt{3}i)^n = 4^n \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

(2)

(iii) Show that

$$\alpha^n + \beta^n + \gamma^n = 2^{2n+1} \cos \frac{2\pi}{3} + \left(-\frac{1}{2} \right)^n$$

where n is an integer.

(3)

(Total 14 marks)

Q8.

(a) (i) Show that $\omega = e^{\frac{2\pi i}{7}}$ is a root of the equation $z^7 = 1$.

(1)

(ii) Write down the five other non-real roots in terms of ω .

(2)

(b) Show that

$$1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 = 0$$

(2)

(c) Show that:

(i) $\omega^2 + \omega^5 = 2 \cos \frac{4\pi}{7};$

(3)

(ii) $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}.$

(4)

(Total 12 marks)

Q9.

(a) Prove by induction that, if n is a positive integer,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

(5)

(b) Hence, given that

$$z = \cos \theta + i \sin \theta$$

show that

$$z^n + \frac{1}{z^n} = 2 \cos n\theta$$

(3)

- (c) Given further that $z + \frac{1}{z} = \sqrt{2}$, find the value of

$$z^{10} + \frac{1}{z^{10}}$$

(4)

(Total 12 marks)

Q10.

- (a) (i) By applying De Moivre's theorem to $(\cos \theta + i \sin \theta)^3$, show that

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$$

(3)

- (ii) Find a similar expression for $\sin 3\theta$.

(1)

- (iii) Deduce that

$$\tan 3\theta = \frac{\tan^3 \theta - 3 \tan \theta}{3 \tan^2 \theta - 1}$$

(3)

- (b) (i) Hence show that $\tan \frac{\pi}{12}$ is a root of the cubic equation

$$x^3 - 3x^2 - 3x + 1 = 0$$

(3)

- (ii) Find two other values of θ , where $0 < \theta < \pi$, for which $\tan \theta$ is a root of this cubic equation.

(2)

- (c) Hence show that

$$\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} = 4$$

(2)

(Total 14 marks)

Q11.

- (a) (i) Expand

$$\left(z + \frac{1}{z}\right) \left(z - \frac{1}{z}\right)$$

(1)

(ii) Hence, or otherwise, expand

$$\left(z + \frac{1}{z}\right)^4 \left(z - \frac{1}{z}\right)^2$$

(3)

(b) (i) Use De Moivre's theorem to show that if $z = \cos \theta + i \sin \theta$ then

$$z^n + \frac{1}{z^n} = 2 \cos n\theta$$

(3)

(ii) Write down a corresponding result for $z^n - \frac{1}{z^n}$.

(1)

(c) Hence express $\cos^4 \theta \sin^2 \theta$ in the form

$$A \cos 6\theta + B \cos 4\theta + C \cos 2\theta + D$$

where A , B , C and D are rational numbers.

(4)

(d) Find $\int \cos^4 \theta \sin^2 \theta \, d\theta$.

(2)

(Total 14 marks)

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Q12.

(a) Prove by induction that, if n is a positive integer,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

(5)

(b) Find the value of $\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6$.

(2)

(c) Show that

$$(\cos \theta + i \sin \theta)(1 + \cos \theta - i \sin \theta) = 1 + \cos \theta + i \sin \theta$$

(3)

(d) Hence show that

$$\left(1 + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^6 = 0$$

(4)

(Total 14 marks)

Q13.

Use De Moivre's Theorem to find the smallest positive angle θ for which

$$(\cos \theta + i \sin \theta)^{15} = -i$$

(Total 5 marks)

Q14.

It is given that $z = e^{i\theta}$.

(a) (i) Show that

$$z + \frac{1}{z} = 2 \cos \theta$$

(2)

(ii) Find a similar expression for

$$z^2 + \frac{1}{z^2}$$

(2)

(iii) Hence show that

$$z^2 - z + 2 - \frac{1}{z} + \frac{1}{z^2} = 4 \cos^2 \theta - 2 \cos \theta$$

(3)

(b) Hence solve the quartic equation

$$z^4 - z^3 + 2z^2 - z + 1 = 0$$

giving the roots in the form $a + ib$.

(5)

(Total 12 marks)